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Summary

- Let A be an $n \times n$ matrix. A is invertible if and only if

- The columns of A span $\mathbb{R}^n$
- For every b in $\mathbb{R}^n$, the system $Ax=b$ is consistent

onto

- The rank of A is n
- The columns of A are linear independent
- The only solution to $Ax=0$ is the zero vector
- The nullity of A is zero
- The reduced row echelon form of A is I_n

One-to-one

- A is a product of elementary matrices
- There exists an $n \times n$ matrix B such that $BA = I_n$
- There exists an $n \times n$ matrix C such that $AC = I_n$

||

square
matrix

57. Let Q be an invertible $n \times n$ matrix. Prove that the subset $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ of R^n is linearly independent if and only if $\{Q\mathbf{u}_1, Q\mathbf{u}_2, \dots, Q\mathbf{u}_k\}$ is linearly independent.

$$\Rightarrow: c_1 Q\mathbf{u}_1 + c_2 Q\mathbf{u}_2 + \dots + c_k Q\mathbf{u}_k = \mathbf{0}$$

$$\Rightarrow Q(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k) = \mathbf{0}$$

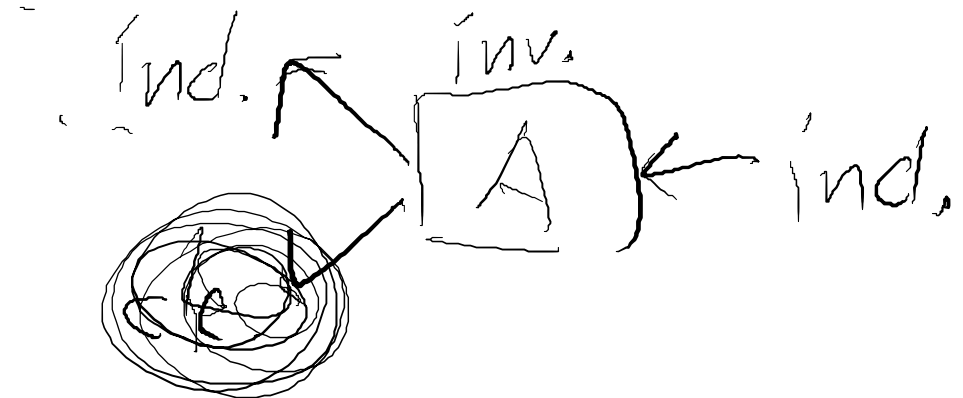
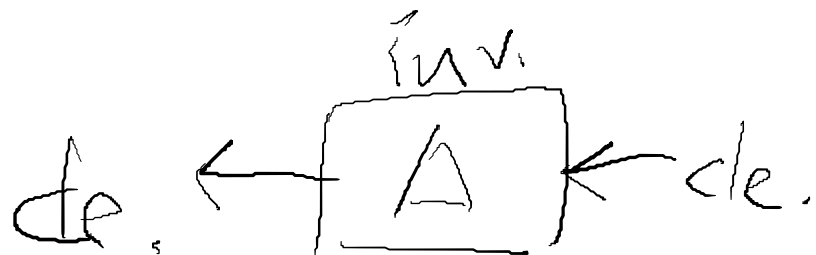
$$Q\mathbf{x} = \mathbf{0}$$

Q is inv.

$$\Rightarrow c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k = \mathbf{0}$$

$$\Rightarrow c_1 = c_2 = \dots = c_k = 0$$

$$\Rightarrow \mathbf{x} = \mathbf{0}$$



62. Prove that if A is an $m \times n$ matrix and B is an $n \times p$ matrix, then $\text{rank}(AB) \leq \text{rank}(B)$.

Hint: Prove that if the k th column of B is not a pivot column of B , then the k th column of AB is not a pivot column of AB .

$$b_k = c_1 b_1 + c_2 b_2 + \dots + c_{k-1} b_{k-1}$$

Ab_k u_k non pivot

$$= c_1 \frac{Ab_1}{u_1} + c_2 \frac{Ab_2}{u_2} + \dots + c_{k-1} \frac{Ab_{k-1}}{u_{k-1}}$$

u_{k-1} B 's no. of pivot

$\geq$ AB 's no. of pivot

$\text{rank } B \geq \text{rank } AB$

66. Let $M = \begin{bmatrix} A & O_1 \\ O_2 & B \end{bmatrix}$, where A and B are square and O_1 and O_2 are zero matrices. Prove that M is invertible if and only if A and B are both invertible.

$\Leftarrow A^{-1}, B^{-1}$ exist

$$\begin{bmatrix} A & O_1 \\ O_2 & B \end{bmatrix} \begin{bmatrix} A^{-1} & O_1 \\ O_2 & B^{-1} \end{bmatrix}$$

$$= \begin{bmatrix} AA^{-1} + O_1 O_2 & AO_1 + O_1 B^{-1} \\ O_2 A^{-1} + BO_2 & O_2 O_1 + BB^{-1} \end{bmatrix}$$

$$\Rightarrow MM^{-1} = I$$

$$M^{-1} = \begin{bmatrix} C & D \\ E & F \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} A & O_1 \\ O_2 & B \end{bmatrix} \begin{bmatrix} C & D \\ E & F \end{bmatrix} = \begin{bmatrix} AC & AD \\ BE & BF \end{bmatrix} = I$$

$$AC = I \Rightarrow C = A^{-1}$$

$$BF = I \Rightarrow F = B^{-1}$$

84. Let A be an $n \times n$ invertible matrix, and let e_j be the j th standard vector of R^n .

- (a) Prove that the j th column of A^{-1} is a solution of $Ax = e_j$.
- (b) Why does the result in (a) imply that A^{-1} is unique?
- (c) Why does the result in (a) imply that $\text{rank } A = n$?

$$(a) \quad Ax = e_j$$

↑

$$A^{-1}e_j$$

$$\underline{A(A^{-1}e_j)} = e_j$$

$$A^{-1} = [a_1' \quad a_2' \quad \dots \quad a_n']$$

↑

$$Ax = e_1$$

↑

$$Ax = e_2$$

↑

$$Ax = e_n$$

90. Let A be an $n \times n$ matrix. Find a property of A that is equivalent to the statement $AB = AC$ if and only if $B = C$

$$\Leftrightarrow A \text{ inv.}$$

$\Leftarrow A$ is inv.

$$AB = AC \Rightarrow \underline{A^{-1}AB} = \underline{A^{-1}AC} \\ \Rightarrow B = C$$

$\Rightarrow$ $\underline{AB = AC \Leftrightarrow B = C}$

$$Au = 0 = A0 \\ \Rightarrow u = 0$$

88. Let A be an $m \times n$ matrix with reduced row echelon form R . Then there exists an invertible matrix P such that $PA = R$ and an invertible matrix Q such that QR^T is the reduced row echelon form of R^T . Describe the matrix PAQ^T in terms of A .

$\text{rank } A = r$

$$\begin{array}{ccccc}
 PAQ^T & = & RQ^T & = & (QR^T)^T \\
 \begin{array}{c} m \times m \\ m \times n \\ n \times n \end{array} & & m \times n & & \begin{array}{c} \uparrow \\ R^T \text{ is } \\ \text{RREF} \end{array}
 \end{array}$$

$A \xrightarrow{m \times n} R \xrightarrow{m \times n} R^T \xrightarrow{n \times m} \text{RREF}(R^T) \xrightarrow{\text{transpose } m \times n} \dots$

$$\left[\begin{array}{c|c|c|c} \text{---} & e_1 & \dots & e_r \end{array} \right] \rightarrow \left[\begin{array}{c} e_1 \\ e_2 \\ \vdots \\ e_r \\ 0 \end{array} \right] \rightarrow \left[\begin{array}{c} e_1 \\ e_2 \\ \vdots \\ e_r \\ 0 \end{array} \right] \rightarrow [e_1 \ e_2 \ \dots \ 0 \ 0 \ 0]$$

86. Let R be an $m \times n$ matrix in reduced row echelon. Find a relationship between the columns of R^T and the columns of $R^T R$.

87. Let R be an $m \times n$ matrix in reduced row echelon form with $\text{rank } R = r$. Prove the following:

(a) The reduced row echelon form of R^T is the $n \times m$ matrix $[\mathbf{e}_1 \quad \dots \quad \mathbf{e}_r \quad \mathbf{0} \quad \dots \quad \mathbf{0}]$, where $\mathbf{e}_j$ is the j th standard vector of R^n for $1 \leq j \leq r$.

(b) $\text{rank } R^T = \text{rank } R$